*Supplement to*

# Plants in limbo — the theory of detrital photosynthesis

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## Supplementary methods

*Halophila ovalis* ramets with several leaf pairs and intact roots were collected from Pelican Point in the brackish Swan Estuary (31.98715°S, 115.82101°E) alongside sediment from adjacent Matilda Bay (31.97572°S, 115.82324°E) on 15th September 2022. I filled the bottom of a 220-L glass flow-through seawater tank with the sediment and planted the ramets to acclimatise them to ambient seawater temperature and salinity and a 12-12-h light-dark cycle (11 ± 1.6 µmol photons $m^{-2}$ $s^{-1}$, mean ± standard deviation, MQ-510, Apogee Instruments Inc., Logan, USA, cf. Masini & Manning 1997) for about two hours. *Amphibolis antarctica* stems with several leaf clusters were collected from fully marine Marmion Lagoon (31.85303°S, 115.75092°E) on 13th October 2022 and immediately used in the experiment. Sediment and seagrasses were collected from less than one metre depth.

The ex situ decomposition experiment was commenced on the respective day of collection by excising individual leaves (*H. ovalis*) or leaf clusters (*A. antarctica*), hereinafter leaves. Leaves were separated from the rest of the plant by cutting the petiole (*H. ovalis*) or stem (*A. antarctica*) just below the leaf base. Excision prematurely induces senescence and was intended to mimic detachment of non-senescent leaves, which may be caused by hydrodynamics or herbivory, rather than abscission. Photosynthesis of six (*H. ovalis*) or nine (*A. antarctica*) samples was immediately measured (see next paragraph). The remaining leaves were enclosed in mesh (3-mm diameter). Mesh bags were weighed down on the sediment with inert aquarium pebbles and kept submerged in the abovementioned 220-L tank for the duration of the experiment. I maintained the 12-h light phase at 11 ± 1.6 µmol photons $m^{-2}$ $s^{-1}$ because this irradiance is sufficient for detrital photosynthesis to play a role (Wright et al. 2024) and is representative of the light environment experienced by detritus in the understorey of kelp forests and seagrass meadows or on deeper sediment flats. To measure detrital photosynthesis, six destructive leaf samples were taken at each of five (*A. antarctica*) to six (*H. ovalis*) timepoints over 34 days following excision.

Light-saturated net photosynthesis ($P_{max}$) of leaves was measured using closed oxygen ($O_2$) incubations (cf. Wright et al. 2024). *A. antarctica* leaves weigh roughly ten times as much as *H. ovalis* leaves (given my data the mass ratio seems to be closer to 14:1). Since I wanted to measure both seagrasses at a comparable leaf mass per volume for the same duration, I incubated ten *H. ovalis* leaves or single *A. antarctica* leaves alongside seawater blanks in sealed glass jars filled with water collected from the flow-through system. Each jar was equipped with a magnetic stir bar and a self-adhesive planar $O_2$ sensor spot (SP-PSt3-SA-NAU-D5-YOP, PreSens Precision Sensing GmbH, Regensburg, Germany) and placed on a magnetic stirrer under a saturating (Masini & Manning 1997, Jamaludin et al. 2006, Said et al. 2021) irradiance of 420 ± 19 µmol photons $m^{-2}$ $s^{-1}$ (Zeus, Ledzeal, Shenzhen Topline Lighting Technology Co. Ltd., Shenzhen, China) in a 20°C room. Care was taken to exclude air from jars. Dissolved $O_2$ (µM) was measured fibre-optically through the glass every 10 s over at least 35 min with a four-channel $O_2$ meter (OXY-4 SMA G2, PreSens Precision Sensing GmbH, Regensburg, Germany). The $O_2$ meter was calibrated using anoxic (1% w/v $Na_2SO_3$) and air-saturated (bubbled with air) ultrapure water. All measurements were corrected for incubation temperature (18 ± 0.73 °C), pressure (1020 ± 6 hPa), and salinity (35 ± 0.48 ‰), respectively using a single temperature dipping probe connected to the $O_2$ meter and placed in a jar filled with seawater, the built-in pressure sensor of the $O_2$ meter, and a handheld refractometer. I blotted and weighed (0.01-g accuracy, 440-33N, Kern & Sohn GmbH, Balingen, Germany) leaves and gravimetrically determined jar volume with ultrapure water to standardise $P_{max}$ as shown in Equation S1.

In the first instance, simple linear models with centred incubation time ($t - \bar{t}$, min) as the predictor and dissolved $O_2$ (μM) as the response variable were fit to measurements from each sample and blank incubation. This yielded posterior probability distributions for slopes ($\beta$, μM $min^{-1}$) which were converted to mass-based $P_{max}$ (μmol $g^{-1}$ $h^{-1}$) as

$$P_{max} = \frac{(\beta_s - \beta_b) \times V \times \Delta t}{m} \quad \text{(S1)}$$

where subscript s and b denote sample and blank incubations from the same measurement group, $V$ is the incubation volume (175 ± 0.17 mL) in litres, m is the sample blotted mass (0.52 ± 0.23 g) in grams and $\Delta t$ is the desired period (1 h) in minutes. I also calculated leaf-based $P_{max}$ because the leaves of *H. ovalis* and leaf clusters *A. antarctica* are quite uniform in size which may lead to less variation in leaf- than mass-based $P_{max}$. For *A. antarctica*, leaf-based $P_{max}$ was simply $(\beta_s - \beta_b) \times V \times \Delta t$. For *H. ovalis*, the rate was additionally divided by 10 because there were 10 leaves in each incubation chamber.

Varying conditions during the 35-min incubations described above could affect $P_{max}$ and confound the experimental effect. Possible confounding variables that I measured are initial incubation $O_2$ (O, μM), mean incubation temperature ($T$, °C), mean incubation pressure ($P$, hPA), incubation salinity ($S$, ‰) and sample mass ($m$, g). Although the oxygen measurements themselves were corrected for temperature, pressure and salinity, photosynthesis could also be influenced by these variables. Similarly, even though $P_{max}$ was standardised by mass (Equation S1), the amount of leaf relative to water in the incubation chamber could additionally affect photosynthesis by changing the availability of carbon and nutrients. I explored these unwanted effects using a multiple linear regression of $P_{max}$ against standardised (minus mean, divided by standard deviation) confounders:

$$\begin{aligned}
P_{max_i} &\sim \text{Normal}(\ \overline{P_{max_i}}\ , \sigma_{P_{max}}\ ) \quad \text{(S2)}\\
\overline{P_{max_i}} &= \alpha + \beta_O O_i + \beta_T T_i + \beta_P P_i + \beta_S S_i + \beta_m m_i\\
\alpha &\sim \text{Normal}(\ 21\ ,\ 10\ )\\
\beta_O, \beta_T, \beta_P, \beta_S, \beta_m &\sim \text{Normal}(\ 0\ ,\ 1\ )\\
\sigma_{P_{max}} &\sim \text{Exponential}(\ 1\ )
\end{aligned}$$

The likelihood for $P_{max}$ is normal because predictions of net photosynthesis can take negative values. Each parameter was stratified by species. Even though $P_{max}$ was only weakly associated with confounders (Figure S1), I decided to account for them in the main model. To predict $P_{max}$ with detrital age ($t$, days), I used the logistic model of detrital photosynthesis (Equation 1) as

$$\begin{aligned}
P_{max_i} &\sim \text{Normal}(\ \overline{P_{max_i}}\ , \sigma_{P_{max}}\ ) \quad \text{(S3)}\\
\overline{P_{max_i}} &= \frac{\alpha_i + \tau_i}{\frac{1 + e^{\frac{5(t_i - \mu_i)}{\mu_i}}}{}} - \tau_i\\
\ln \alpha_i &= \overline{\ln \alpha}\ + \beta_O O_i + \beta_T T_i + \beta_P P_i + \beta_S S_i + \beta_m m_i\\
\ln \mu_i &= \overline{\ln \mu}\ + \gamma_O O_i + \gamma_T T_i + \gamma_P P_i + \gamma_S S_i + \gamma_m m_i\\
\ln \tau_i &= \overline{\ln \tau}\ + \delta_O O_i + \delta_T T_i + \delta_P P_i + \delta_S S_i + \delta_m m_i\\
\overline{\ln \alpha} &\sim \text{Normal}(\ \ln 21\ ,\ 0.4\ )\\
\overline{\ln \mu} &\sim \text{Normal}(\ \ln 14\ ,\ 0.6\ )\\
\overline{\ln \tau} &\sim \text{Normal}(\ \ln 2\ ,\ 0.4\ )\\
\beta_O, \beta_T, \beta_P, \beta_S, \beta_m, \gamma_O, \gamma_T, \gamma_P, \gamma_S, \gamma_m, \delta_O, \delta_T, \delta_P, \delta_S, \delta_m &\sim \text{Normal}(\ 0\ ,\ 1\ )\\
\sigma_{P_{max}} &\sim \text{Exponential}(\ 1\ )
\end{aligned}$$

where $\alpha$ is baseline photosynthesis (μmol $g^{-1}$ $h^{-1}$), $\mu$ is the photosynthetic half-life (days) and $\tau$ is respiration of dead leaves (μmol $g^{-1}$ $h^{-1}$). These parameters are strictly positive, so their logarithm is described by a multiple linear regression against the confounders (cf. Equation S2). Some species-specific influence of confounders on $\alpha$ and $\mu$ is evident (Figure S1), so their inclusion in this model is important for accurate prediction. I also ran this model for leaf-based $P_{max}$, adjusting the priors for $\alpha$ and $\tau$ accordingly.

Prior probability distributions for parameters were chosen based on published estimates. For the initial linear regressions of $O_2$ against incubation time, I centred the intercept prior on regional average seawater $O_2$ and the slope prior on zero. For $\alpha$ (Equation S2, S3), I chose the prior mean based on various published $P_{max}$ estimates for *H. ovalis* (40 ± 57 µmol $g^{-1}$ $h^{-1}$, 1.8 ± 2.5 µmol $leaf^{-1}$ $h^{-1}$, n = 32, Björk et al. 1997, Jamaludin et al. 2006, Borum et al. 2016, Lamit & Tanaka 2021, Said et al. 2021, 2026) and A. *antarctica* (14 ± 9 µmol $g^{-1}$ $h^{-1}$, 8.8 ± 5.5 µmol $leaf^{-1}$ $h^{-1}$, n = 8, Masini & Manning 1997, Borum et al. 2016, Said et al. 2026). Units were converted where necessary by multiplying by mean dry-fresh mass ratios for *H. ovalis* (0.21 g dry mass $g^{-1}$ fresh mass) and A. *antarctica* (0.29 g dry mass $g^{-1}$ fresh mass) (de los Santos et al. 2012, Borum et al. 2016) and my own leaf fresh masses for *H. ovalis* (0.04 ± 0.01 g $leaf^{-1}$) and A. *antarctica* (0.61 ± 0.27 g $leaf^{-1}$). For $\mu$ (Equation S3), I settled on a median of 14 days, close to the minimum of 15 days for *Ecklonia radiata* (Wright et al. 2024) because I expected photosynthetic decay to be faster than for seaweeds. For $\tau$ (Equation S3), I centred the prior around published estimates of $O_2$ consumption by leaf litter of *Zostera marina* (17 ± 5.6 µmol $g^{-1}$ $h^{-1}$, n = 16, Blum & Mills 1991) and *Posidonia oceanica* (1.3 ± 1.3 µmol $g^{-1}$ $h^{-1}$, n = 46, Mateo & Romero 1996). Units were again converted by multiplying by mean dry-fresh mass ratios for *Z. marina* (0.23 g dry mass $g^{-1}$ fresh mass, Evans et al. 1986) and *P. oceanica* (0.24 g dry mass $g^{-1}$ fresh mass, Apostolaki et al. 2024). I chose a reasonable standard deviation for each prior based on prior simulation and variation in the published estimates.

The treatment of measurement error deserves brief mention. $\beta$ and $V$ (Equation S1) and hence $P_{max}$ (Equation S1, S2, S3) as well as O and $T$ (Equation S2, S3) are measured with error, which I propagated into each downstream model as the standard deviation of upstream posteriors. This was achieved by parameterising latent variables for $P_{max}$, O and $T$. In all cases I used the error structure mean ~ Normal( latent , standard deviation ) where each observed mean is described by a normal distribution centred on the unobserved true value with observed error (McElreath 2020). For the response variable $P_{max}$, the latent variable is described by the normal likelihood (Equation S2, S3). For the standardised predictor variables O and $T$, the latent variable is described by a standard normal prior: latent ~ Normal( 0 , 1 ). Wherever a variable is measured with error, I visualised each observation as a univariate or bivariate distribution (violin or ellipse) rather than a point.

Data analysis and visualisation were performed in R v4.5.1 (R Core Team 2026) within the integrated development environment RStudio v2026.01.0+392 (RStudio Team 2026). General workflow was facilitated by the tidyverse package family v2.0.0 (Wickham et al. 2019) and here v1.0.2 (Müller 2025). Hamiltonian Monte Carlo models were written in Stan (Carpenter et al. 2017) and run with CmdStan v2.17.1 (Lee et al. 2026) via the R interface cmdstanr v0.9.0 (Gabry et al. 2025). All models were run with 8 Markov chains spread across all cores with $10^4$ warmup and sampling iterations each. Convergence and smooth sampling were optimised by assessing $\hat{R}$ scores and visually scrutinising trace rank and pair plots with bayesplot v1.14.0 (Gabry et al. 2019). All reported results are posteriors and derived central tendencies and intervals calculated with tidybayes v3.0.7 (Kay 2024). Figure and table creation was facilitated by ggridges v0.5.7 (Wilke 2021), ggh4x v0.3.1 (van den Brand 2025), geomtextpath v0.2.0 (Cameron & van den Brand 2022), patchwork v1.3.2 (Pedersen 2025) and officer v0.7.0 (Gohel et al. 2025). For details, all data, code, diagnostic plots and priors and posteriors can be consulted at github.com/lukaseamus/plant-limbo and doi.org/10.5281/zenodo.23182864. Illustrations were made with Affinity Designer v1.10.8 (Serif Ltd., Nottingham, UK).

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## Supplementary figures

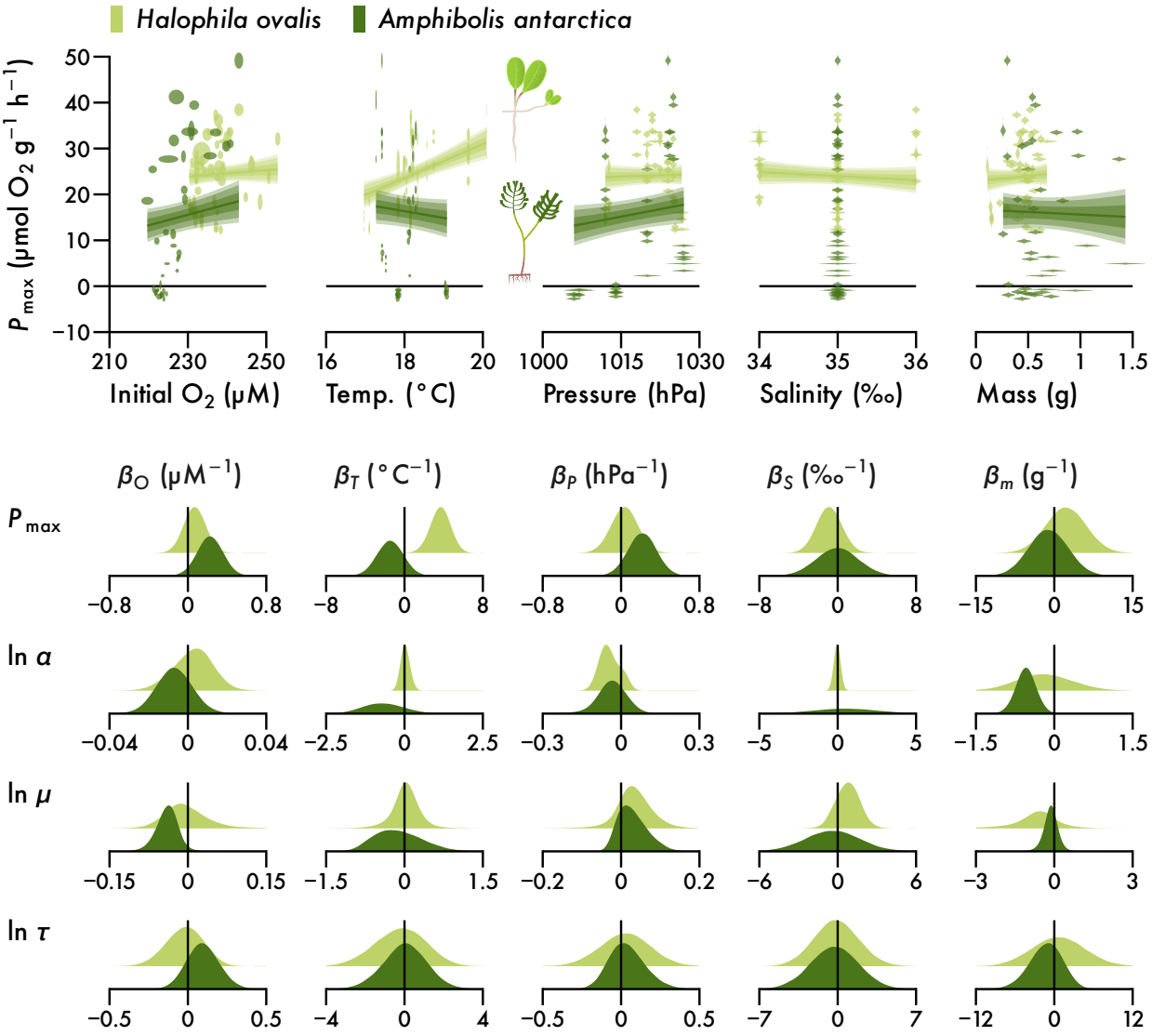



**Figure S1**. Effect of confounding variables on light-saturated net photosynthesis ($P_{max}$) of excised *Halophila ovalis* and *Amphibolis antarctica* leaves. $P_{max}$ is standardised by leaf fresh mass. Ellipses and violins are kernel density estimates of observations with measurement error (n = 42 for *H. ovalis*, n = 39 for *A. antarctica*). Lines and intervals are medians and the central 50%, 80% and 90% of posterior probability for predicted means. Distributions are kernel density estimates of slope posteriors (Equation S2, S3).

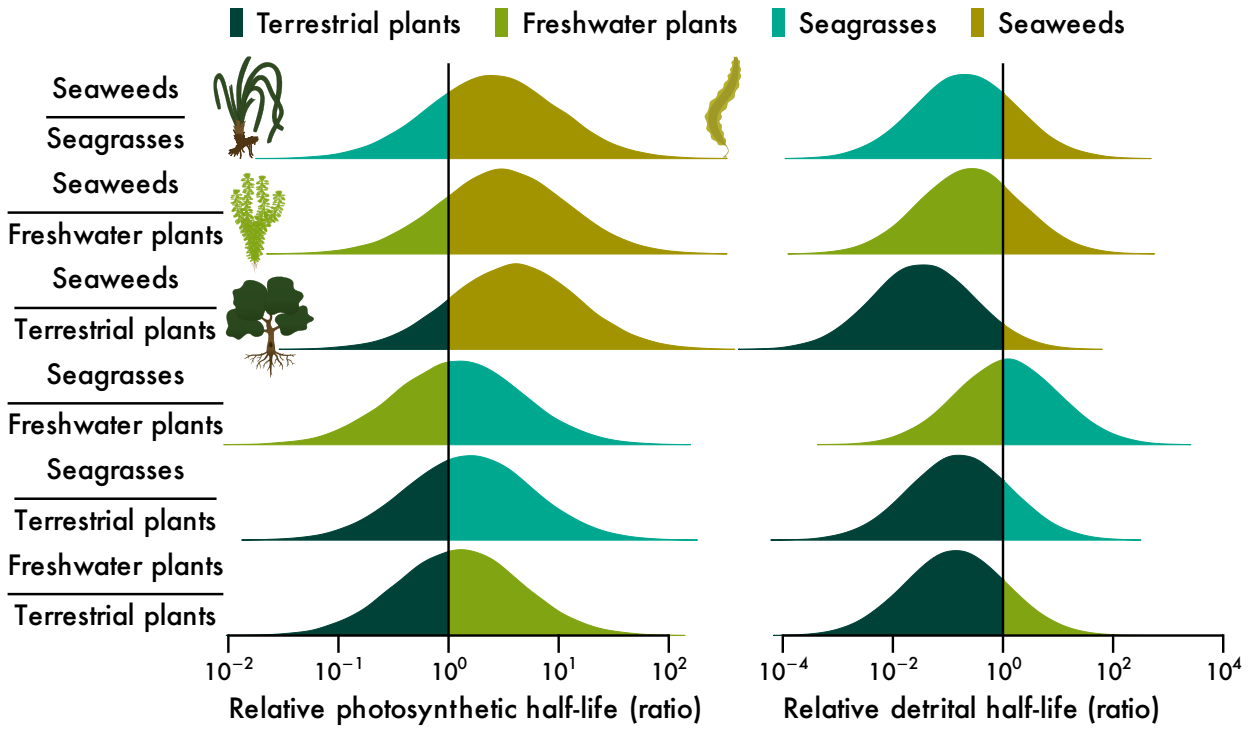



**Figure S2**. Photosynthetic and detrital half-life contrasts between plant groups. The group with more probability (coloured area) either side of the identity line at ratio = 1 has a greater half-life. Distributions are kernel density estimates of posterior ratios.

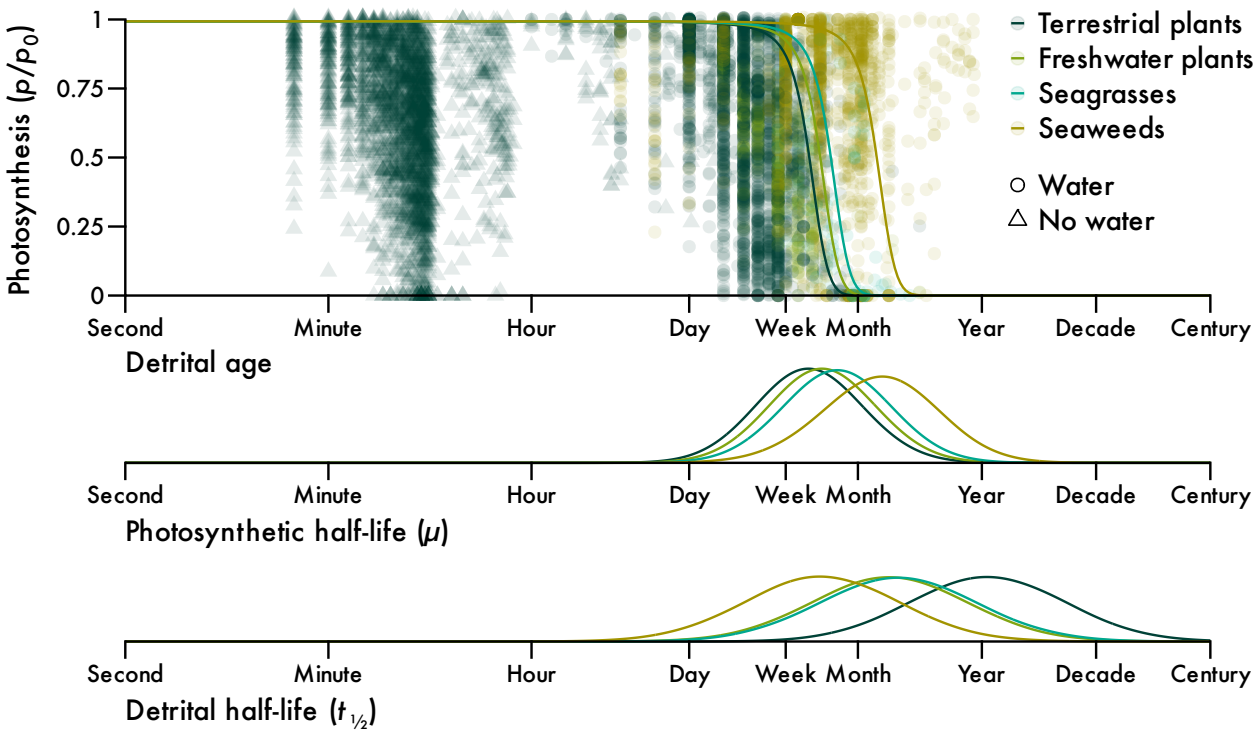



**Figure S3**. Persistence of photosynthesis in detritus with and without access to water (cf. Figure 3). For comparison, the exponential half-life of detrital mass is shown alongside the photosynthetic half-life. Lines are median predictions of normalised photosynthesis in the top panel and kernel density estimates of posterior probability distributions in the middle and bottom panels (cf. Figure 3). Note the log scale.

**Supplementary tables**

**Table S1**. Species included in the meta-analysis. Phyla are given in bold and orders are underlined. Seaweeds are comprised of Chlorophyta and Heterokontophyta. Seagrasses are marked with an asterisk. All remaining Alismatales are freshwater plants and the rest of Streptophyta is terrestrial. The longest experimental durations and largest numbers of studies, experiments and observations in each non-taxonomic group are highlighted in bold.

| | Species | Days | Studies | Exper. | Observ. |
|---|---|---|---|---|---|
| **Chlorophyta** | | | | | |
| Ulvales | | | | | |
| Ulvaceae | *Ulva clathrata* | 30 | 1 | 1 | 6 |
| | *Ulva intestinalis* | 30 | 1 | 1 | 6 |
| | *Ulva reticulata* | 30 | 1 | 1 | 6 |
| **Heterokontophyta** | | | | | |
| Desmarestiales | | | | | |
| Desmarestiaceae | *Desmarestia anceps* | **313** | 1 | 1 | 65 |
| Fucales | | | | | |
| Durvillaeaceae | *Durvillaea antarctica* | 14 | 2 | 16 | 448 |
| Sargassaceae | *Sargassum fallax* | 14 | 1 | 9 | 84 |
| | *Sargassum spinuligerum* | 14 | 1 | 13 | 159 |
| Seirococcaceae | *Scytothalia dorycarpa* | 14 | 1 | 6 | 48 |
| Laminariales | | | | | |
| Laminariaceae | *Laminaria digitata* | 46 | 4 | 6 | 191 |
| | *Laminaria hyperborea* | 168 | **7** | **56** | **559** |
| | *Laminaria ochroleuca* | 56 | 4 | 51 | 416 |
| | *Macrocystis pyrifera* | 55 | 4 | 23 | 375 |
| | *Saccharina latissima* | 39 | 2 | 2 | 55 |
| Lessoniaceae | *Ecklonia radiata* | 119 | 2 | 10 | 117 |
| Tilopteridales | | | | | |
| Phyllariaceae | *Saccorhiza polyschides* | 39 | 1 | 1 | 38 |
| **Streptophyta** | | | | | |
| Alismatales | | | | | |
| Cymodoceaceae | *Amphibolis antarctica** | 34 | **1** | 1 | 39 |
| Hydrocharitaceae | *Elodea densa* | 16 | 2 | 3 | 43 |
| | *Halophila ovalis** | 34 | **1** | 1 | 42 |
| | *Hydrilla verticillata* | 12 | **7** | **16** | **506** |
| | *Thalassia testudinum** | 52 | **1** | 1 | 14 |
| | *Vallisneria americana* | **104** | 1 | 1 | 5 |
| | *Vallisneria spiralis* | 9 | 6 | 6 | 59 |
| Potamogetonaceae | *Potamogeton* sp. | **104** | 1 | 1 | 5 |
| | *Stuckenia pectinata* | 9 | 5 | 5 | 55 |
| Zosteraceae | *Zostera marina** | **85** | **1** | 1 | 7 |
| | *Zostera muelleri** | 16 | **1** | **4** | **58** |
| Apiales | | | | | |
| Apiaceae | *Coriandrum sativum* | 9 | 1 | 1 | 24 |
| | *Petroselinum crispum* | 7 | 1 | 1 | 7 |
| Asterales | | | | | |
| Asteraceae | *Chrysanthemum × morifolium* | 12 | 1 | 2 | 30 |
| | *Helianthus annuus* | 0.024 | 1 | 1 | 36 |
| | *Helianthus tuberosus* | 0.026 | 1 | 1 | 36 |
| | *Inula racemosa* | 0.025 | 1 | 1 | 36 |

| | Species | Days | Studies | Exper. | Observ. |
|---|---|---|---|---|---|
| | *Smallanthus connatus* | 0.026 | 1 | 1 | 36 |
| Brassicales | | | | | |
| Brassicaceae | *Arabidopsis thaliana* | 6 | 2 | 3 | 60 |
| | *Brassica napus* | 1.7 | 1 | 25 | 125 |
| | *Brassica oleracea* | 10 | 1 | 3 | 22 |
| | *Nasturtium officinale* | 5 | 2 | 3 | 39 |
| | *Raphanus raphanistrum* | 6 | 1 | 4 | 20 |
| | *Sinapis alba* | 6 | 1 | 4 | 20 |
| Tropaeolaceae | *Tropaeolum majus* | 8 | 2 | 3 | 27 |
| Caryophyllales | | | | | |
| Amaranthaceae | *Amaranthus cruentus* | 0.026 | 1 | 1 | 36 |
| | *Celosia argentea* | 0.024 | 1 | 1 | 36 |
| | *Chenopodium berlandieri* | 0.27 | 1 | 2 | 26 |
| | *Gomphrena serrata* | 0.024 | 1 | 1 | 36 |
| | *Spinacia oleracea* | 10 | 3 | 4 | 85 |
| Polygonaceae | *Koenigia weyrichii* | 0.028 | 1 | 1 | 36 |
| Cucurbitales | | | | | |
| Cucurbitaceae | *Cucumis sativus* | 13 | 2 | 3 | 13 |
| | *Sicyos edulis* | 10 | 1 | 2 | 48 |
| Ericales | | | | | |
| Theaceae | *Camellia sinensis* | 4 | 1 | 2 | 11 |
| Fabales | | | | | |
| Fabaceae | *Cercis canadensis* | 0.0052 | 1 | 4 | 576 |
| | *Lathyrus oleraceus* | 17 | 2 | 2 | 42 |
| | *Phaseolus vulgaris* | 2.1 | 1 | 1 | 10 |
| | *Trifolium subterraneum* | 7 | 2 | 5 | 25 |
| | *Vigna unguiculata* | 4 | 1 | 10 | 170 |
| Fagales | | | | | |
| Betulaceae | *Betula pendula* | 0.029 | 1 | 1 | 36 |
| Fagaceae | *Quercus alba* | 0.01 | 1 | 1 | 24 |
| | *Quercus muehlenbergii* | 0.0052 | 1 | 1 | 192 |
| | *Quercus robur* | 0.024 | 2 | 2 | 228 |
| | *Quercus rubra* | 0.01 | 1 | 1 | 24 |
| Juglandaceae | *Juglans nigra* | 0.01 | 1 | 1 | 24 |
| Lamiales | | | | | |
| Lamiaceae | *Salvia officinalis* | 11 | 1 | 1 | 8 |
| Laurales | | | | | |
| Lauraceae | *Cinnamomum tamala* | 21 | 1 | 1 | 36 |
| Pinales | | | | | |
| Pinaceae | *Larix sibirica* | 0.025 | 1 | 1 | 36 |
| | *Pinus sibirica* | 0.023 | 1 | 1 | 33 |
| | *Pinus sylvestris* | 0.022 | 1 | 1 | 33 |
| Poales | | | | | |
| Cyperaceae | *Bolboschoenus fluviatilis* | **104** | 1 | 1 | 5 |
| Poaceae | *Avena sativa* | 8 | 8 | 28 | 161 |
| | *Hordeum vulgare* | 11 | **13** | 25 | 466 |
| | *Lolium multiflorum* | 5 | 1 | 3 | 44 |
| | *Lolium pratense* | 7 | 5 | 8 | 112 |
| | *Oryza sativa* | 27 | 11 | **48** | 327 |
| | *Panicum miliaceum* | 5 | 2 | 2 | 69 |
| | *Secale cereale* | 5 | 2 | 9 | 102 |

| | Species | Days | Studies | Exper. | Observ. |
|---|---|---|---|---|---|
| | *Triticum aestivum* | 10 | **13** | 41 | 646 |
| | *Triticum turgidum* | 6 | 1 | 1 | 7 |
| | *Triticum vulgare* | 6 | 1 | 2 | 48 |
| | *Zea mays* | 6 | 3 | 4 | 58 |
| Typhaceae | *Typha angustifolia* | **104** | 1 | 1 | 5 |
| Polypodiales | | | | | |
| Nephrolepidaceae | *Nephrolepis exaltata* | 10 | 1 | 4 | 24 |
| Rosales | | | | | |
| Rosaceae | *Fragaria × ananassa* | 0.0056 | 1 | 1 | 663 |
| | *Malus domestica* | 21 | 3 | 6 | 150 |
| | *Malus hupehensis* | 15 | 1 | 1 | 18 |
| Sapindales | | | | | |
| Rutaceae | *Citrus × aurantium* | 10 | 1 | 1 | 5 |
| Sapindaceae | *Acer saccharinum* | 0.25 | 1 | 4 | 36 |
| | *Acer truncatum* | 0.0052 | 1 | 1 | 192 |
| Solanales | | | | | |
| Convolvulaceae | *Ipomoea batatas* | 3 | 1 | 2 | 24 |
| Solanaceae | *Nicotiana rustica* | 10 | 1 | 2 | 180 |
| | *Nicotiana tabacum* | 9 | 1 | 2 | 24 |
| | *Solanum lycopersicum* | 5 | 1 | 6 | 192 |
| | *Solanum melongena* | 16 | 1 | 1 | 24 |
| | *Solanum tuberosum* | 0.026 | 1 | 1 | 36 |
| Vitales | | | | | |
| Vitaceae | *Vitis vinifera* | 0.0052 | 1 | 5 | **1280** |

**Table S2**. Predictions of photosynthetic half-life (days, $\mu$ in Equation 2) for new experiments on unobserved species in each plant group in this study as well as new plant groups. Estimates are stratified by observed differences in light condition and the type of response variable. All estimates are given as median (ln mean ± standard deviation).

| Light | Response | Photosynthetic half-life |
|---|---|---|
| **Terrestrial plants** | | |
| Light | Photosynthesis | 13 (2.6 ± 0.94) |
| | Chlorophyll | 18 (2.9 ± 0.93) |
| Dark | Photosynthesis | 7.2 (2 ± 0.92) |
| | Chlorophyll | 9.5 (2.3 ± 0.92) |
| **Freshwater plants** | | |
| Light | Chlorophyll | 19 (2.9 ± 0.98) |
| Dark | Photosynthesis | 11 (2.4 ± 0.95) |
| | Chlorophyll | 16 (2.8 ± 0.95) |
| **Seagrasses** | | |
| Light | Photosynthesis | 19 (2.9 ± 0.98) |
| | Chlorophyll | 26 (3.3 ± 0.98) |
| Dark | Chlorophyll | 16 (2.8 ± 1) |
| **Seaweeds** | | |
| Light | Photosynthesis | 51 (3.9 ± 0.94) |
| | Chlorophyll | 81 (4.4 ± 0.96) |
| Dark | Photosynthesis | 24 (3.2 ± 1) |
| **All plants** | | |
| Light | Photosynthesis | 17 (2.8 ± 1.1) |
| | Chlorophyll | 24 (3.2 ± 1.1) |
| Dark | Photosynthesis | 12 (2.4 ± 1.1) |
| | Chlorophyll | 16 (2.8 ± 1.1) |

**Table S3**. Predictions of photosynthetic half-life (days, $\mu$ in Equation 2) for new experiments on each species in this study (cf. Table S1 for complete taxonomy). Estimates are stratified by observed differences in light condition and the type of response variable. All estimates are given as median (ln mean ± standard deviation).

| Species | Light | Response | Photosynthetic half-life |
|---|---|---|---|
| **Seaweeds** | | | |
| *Desmarestia anceps* | Light | Chlorophyll | 154 (5 ± 0.9) |
| *Durvillaea antarctica* | Light | Photosynthesis | 59 (4.1 ± 0.87) |
| | | Chlorophyll | 95 (4.6 ± 0.9) |
| *Ecklonia radiata* | Light | Photosynthesis | 32 (3.5 ± 0.78) |
| | Dark | Photosynthesis | 15 (2.7 ± 0.87) |
| *Laminaria digitata* | Light | Photosynthesis | 68 (4.2 ± 0.86) |
| | | Chlorophyll | 109 (4.7 ± 0.88) |
| *Laminaria hyperborea* | Light | Photosynthesis | 62 (4.1 ± 0.73) |
| | | Chlorophyll | 99 (4.6 ± 0.79) |
| *Laminaria ochroleuca* | Light | Photosynthesis | 68 (4.2 ± 0.74) |
| | | Chlorophyll | 108 (4.7 ± 0.79) |
| *Macrocystis pyrifera* | Light | Photosynthesis | 60 (4.1 ± 0.84) |
| | | Chlorophyll | 96 (4.6 ± 0.86) |
| *Saccharina latissima* | Light | Photosynthesis | 60 (4.1 ± 0.9) |
| | | Chlorophyll | 96 (4.6 ± 0.91) |
| *Saccorhiza polyschides* | Light | Chlorophyll | 87 (4.5 ± 0.92) |

| Species | Light | Response | Photosynthetic half-life |
|---|---|---|---|
| *Sargassum fallax* | Light | Photosynthesis | 30 (3.4 ± 0.82) |
| *Sargassum spinuligerum* | Light | Photosynthesis | 64 (4.2 ± 0.86) |
| | | Chlorophyll | 102 (4.6 ± 0.89) |
| *Scytothalia dorycarpa* | Light | Photosynthesis | 45 (3.8 ± 0.95) |
| *Ulva clathrata* | Light | Chlorophyll | 85 (4.4 ± 0.94) |
| *Ulva intestinalis* | Light | Chlorophyll | 84 (4.4 ± 0.94) |
| *Ulva reticulata* | Light | Chlorophyll | 85 (4.4 ± 0.94) |
| **Seagrasses** | | | |
| *Amphibolis antarctica* | Light | Photosynthesis | 14 (2.6 ± 0.88) |
| *Halophila ovalis* | Light | Photosynthesis | 27 (3.3 ± 0.91) |
| *Thalassia testudinum* | Light | Chlorophyll | 26 (3.2 ± 0.87) |
| *Zostera marina* | Light | Chlorophyll | 26 (3.3 ± 0.87) |
| *Zostera muelleri* | Light | Chlorophyll | 25 (3.2 ± 0.86) |
| | Dark | Chlorophyll | 16 (2.8 ± 0.86) |
| **Freshwater plants** | | | |
| *Elodea densa* | Light | Chlorophyll | 18 (2.9 ± 0.89) |
| | Dark | Chlorophyll | 15 (2.7 ± 0.86) |
| *Hydrilla verticillata* | Light | Chlorophyll | 14 (2.6 ± 0.79) |
| | Dark | Photosynthesis | 7.7 (2 ± 0.78) |
| | Dark | Chlorophyll | 12 (2.5 ± 0.78) |
| *Potamogeton sp.* | Dark | Chlorophyll | 25 (3.2 ± 0.88) |
| *Stuckenia pectinata* | Dark | Photosynthesis | 7.1 (2 ± 0.8) |
| | Dark | Chlorophyll | 11 (2.4 ± 0.82) |
| *Vallisneria americana* | Dark | Chlorophyll | 25 (3.2 ± 0.88) |
| *Vallisneria spiralis* | Dark | Photosynthesis | 7.8 (2.1 ± 0.79) |
| | Dark | Chlorophyll | 12 (2.5 ± 0.81) |
| **Terrestrial plants** | | | |
| *Acer saccharinum* | Light | Photosynthesis | 13 (2.6 ± 0.93) |
| *Acer truncatum* | Light | Photosynthesis | 13 (2.6 ± 0.94) |
| *Amaranthus cruentus* | Light | Photosynthesis | 13 (2.6 ± 0.93) |
| *Arabidopsis thaliana* | Dark | Photosynthesis | 4.3 (1.5 ± 0.81) |
| | Dark | Chlorophyll | 5.8 (1.8 ± 0.8) |
| *Avena sativa* | Light | Chlorophyll | 11 (2.4 ± 0.78) |
| | Dark | Chlorophyll | 5.8 (1.8 ± 0.76) |
| *Betula pendula* | Light | Photosynthesis | 13 (2.6 ± 0.94) |
| *Bolboschoenus fluviatilis* | Dark | Chlorophyll | 20 (3 ± 0.9) |
| *Brassica napus* | Light | Chlorophyll | 18 (2.9 ± 0.91) |
| *Brassica oleracea* | Dark | Chlorophyll | 8.9 (2.2 ± 0.85) |
| *Camellia sinensis* | Light | Photosynthesis | 14 (2.7 ± 0.91) |
| *Celosia argentea* | Light | Photosynthesis | 13 (2.6 ± 0.93) |
| *Cercis canadensis* | Light | Photosynthesis | 13 (2.6 ± 0.93) |
| *Chenopodium berlandieri* | Light | Photosynthesis | 13 (2.6 ± 0.94) |
| *Chrysanthemum × morifolium* | Dark | Chlorophyll | 9.4 (2.2 ± 0.82) |
| *Cinnamomum tamala* | Dark | Chlorophyll | 9.6 (2.3 ± 0.84) |
| *Citrus × aurantium* | Light | Chlorophyll | 19 (2.9 ± 0.9) |
| *Coriandrum sativum* | Dark | Chlorophyll | 9.6 (2.3 ± 0.87) |
| *Cucumis sativus* | Light | Chlorophyll | 19 (3 ± 0.89) |
| | Dark | Chlorophyll | 10 (2.3 ± 0.88) |
| *Fragaria × ananassa* | Light | Photosynthesis | 13 (2.6 ± 0.93) |
| *Gomphrena serrata* | Light | Photosynthesis | 13 (2.6 ± 0.93) |
| *Helianthus annuus* | Light | Photosynthesis | 13 (2.6 ± 0.94) |

| Species | Light | Response | Photosynthetic half-life |
|---|---|---|---|
| *Helianthus tuberosus* | Light | Photosynthesis | 13 (2.6 ± 0.93) |
| *Hordeum vulgare* | Light | Photosynthesis | 9.2 (2.2 ± 0.77) |
| | | Chlorophyll | 12 (2.5 ± 0.76) |
| | Dark | Photosynthesis | 5 (1.6 ± 0.75) |
| | | Chlorophyll | 6.6 (1.9 ± 0.75) |
| *Inula racemosa* | Light | Photosynthesis | 13 (2.6 ± 0.94) |
| *Ipomoea batatas* | Dark | Photosynthesis | 7.9 (2.1 ± 0.9) |
| | | Chlorophyll | 11 (2.4 ± 0.9) |
| *Juglans nigra* | Light | Photosynthesis | 13 (2.6 ± 0.93) |
| *Koenigia weyrichii* | Light | Photosynthesis | 13 (2.6 ± 0.93) |
| *Larix sibirica* | Light | Photosynthesis | 13 (2.6 ± 0.94) |
| *Lathyrus oleraceus* | Light | Photosynthesis | 16 (2.8 ± 0.89) |
| | | Chlorophyll | 22 (3.1 ± 0.88) |
| *Lolium multiflorum* | Light | Photosynthesis | 12 (2.5 ± 0.92) |
| | | Chlorophyll | 16 (2.8 ± 0.92) |
| *Lolium pratense* | Dark | Photosynthesis | 4.1 (1.4 ± 0.77) |
| | | Chlorophyll | 5.5 (1.7 ± 0.77) |
| *Malus domestica* | Light | Photosynthesis | 24 (3.2 ± 0.86) |
| | | Chlorophyll | 33 (3.5 ± 0.85) |
| | Dark | Photosynthesis | 13 (2.6 ± 0.85) |
| | | Chlorophyll | 18 (2.9 ± 0.85) |
| *Malus hupehensis* | Dark | Chlorophyll | 13 (2.6 ± 0.88) |
| *Nasturtium officinale* | Dark | Chlorophyll | 4.3 (1.5 ± 0.81) |
| *Nephrolepis exaltata* | Light | Photosynthesis | 18 (2.9 ± 0.87) |
| | | Chlorophyll | 24 (3.2 ± 0.86) |
| | Dark | Photosynthesis | 9.8 (2.3 ± 0.86) |
| | | Chlorophyll | 13 (2.6 ± 0.85) |
| *Nicotiana rustica* | Dark | Photosynthesis | 8.5 (2.1 ± 0.84) |
| | | Chlorophyll | 11 (2.4 ± 0.83) |
| *Nicotiana tabacum* | Light | Photosynthesis | 16 (2.8 ± 0.89) |
| | | Chlorophyll | 21 (3.1 ± 0.88) |
| *Oryza sativa* | Light | Photosynthesis | 10 (2.4 ± 0.77) |
| | | Chlorophyll | 14 (2.6 ± 0.76) |
| | Dark | Photosynthesis | 5.7 (1.7 ± 0.76) |
| | | Chlorophyll | 7.5 (2 ± 0.75) |
| *Panicum miliaceum* | Dark | Chlorophyll | 5.4 (1.7 ± 0.82) |
| *Petroselinum crispum* | Dark | Chlorophyll | 10 (2.3 ± 0.89) |
| *Phaseolus vulgaris* | Light | Photosynthesis | 13 (2.6 ± 0.93) |
| *Pinus sibirica* | Light | Photosynthesis | 13 (2.6 ± 0.93) |
| *Pinus sylvestris* | Light | Photosynthesis | 13 (2.6 ± 0.94) |
| *Quercus alba* | Light | Photosynthesis | 13 (2.6 ± 0.94) |
| *Quercus muehlenbergii* | Light | Photosynthesis | 13 (2.6 ± 0.93) |
| *Quercus robur* | Light | Photosynthesis | 13 (2.6 ± 0.93) |
| *Quercus rubra* | Light | Photosynthesis | 13 (2.6 ± 0.94) |
| *Raphanus raphanistrum* | Light | Photosynthesis | 14 (2.7 ± 0.89) |
| | | Chlorophyll | 19 (3 ± 0.89) |
| *Salvia officinalis* | Dark | Chlorophyll | 12 (2.5 ± 0.89) |
| *Secale cereale* | Light | Photosynthesis | 13 (2.6 ± 0.86) |
| | | Chlorophyll | 18 (2.9 ± 0.85) |
| | Dark | Photosynthesis | 7.2 (2 ± 0.85) |
| | | Chlorophyll | 9.5 (2.3 ± 0.85) |

| Species | Light | Response | Photosynthetic half-life |
|---|---|---|---|
| *Sicyos edulis* | Dark | Photosynthesis | 7.4 (2 ± 0.82) |
| | | Chlorophyll | 9.9 (2.3 ± 0.82) |
| *Sinapis alba* | Light | Photosynthesis | 15 (2.7 ± 0.89) |
| | | Chlorophyll | 20 (3 ± 0.89) |
| *Smallanthus connatus* | Light | Photosynthesis | 13 (2.6 ± 0.93) |
| *Solanum lycopersicum* | Light | Photosynthesis | 11 (2.4 ± 0.85) |
| *Solanum melongena* | Light | Photosynthesis | 15 (2.7 ± 0.88) |
| *Solanum tuberosum* | Light | Photosynthesis | 13 (2.6 ± 0.93) |
| *Spinacia oleracea* | Dark | Photosynthesis | 5.8 (1.8 ± 0.8) |
| | | Chlorophyll | 7.7 (2 ± 0.79) |
| *Trifolium subterraneum* | Light | Photosynthesis | 12 (2.5 ± 0.88) |
| | | Chlorophyll | 16 (2.8 ± 0.88) |
| *Triticum aestivum* | Light | Photosynthesis | 5.5 (1.7 ± 0.75) |
| | | Chlorophyll | 7.3 (2 ± 0.75) |
| | Dark | Photosynthesis | 3 (1.1 ± 0.74) |
| | | Chlorophyll | 4 (1.4 ± 0.74) |
| *Triticum turgidum* | Light | Chlorophyll | 18 (2.9 ± 0.92) |
| *Triticum vulgare* | Light | Photosynthesis | 13 (2.6 ± 0.88) |
| | Dark | Photosynthesis | 7.3 (2 ± 0.87) |
| *Tropaeolum majus* | Light | Chlorophyll | 20 (3 ± 0.87) |
| | Dark | Chlorophyll | 11 (2.4 ± 0.87) |
| *Typha angustifolia* | Dark | Chlorophyll | 22 (3.1 ± 0.91) |
| *Vigna unguiculata* | Dark | Photosynthesis | 5.3 (1.7 ± 0.78) |
| | | Chlorophyll | 7.2 (2 ± 0.78) |
| *Vitis vinifera* | Light | Photosynthesis | 13 (2.6 ± 0.93) |
| *Zea mays* | Light | Photosynthesis | 14 (2.6 ± 0.91) |
| | | Chlorophyll | 19 (2.9 ± 0.9) |
| | Dark | Chlorophyll | 10 (2.3 ± 0.89) |

**Supplementary references**

Studies included in the meta-analysis in addition to the present study. Note that 130 references are listed because the data from three studies were not independent and thus merged.


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